LUIS HUMBERTO MOTA REYES

Mechanical Engineering Student · Mechanical Design, Structures & Aerospace Manufacturing

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PROFILE

Mechanical Engineering student and **Certified SOLIDWORKS Professional (CSWP)** leading the Design & Manufacturing team of Ignitia, Tec de Monterrey's experimental rocketry team. Hands-on experience taking aluminum and composite components from CAD and FEA/CFD simulation to manufacturing, assembly and static fire testing. Former captain of Mexico's 2025 FRC national champion team. Seeking internships in the aerospace and mechanical industry.

EDUCATION

Tecnológico de Monterrey, Campus Guadalajara — Expected Graduation: July 2029
B.S. in Mechanical Engineering

CERTIFICATIONS

- **Certified SOLIDWORKS Professional (CSWP)** — Dassault Systèmes, Aug 2026 · Credential ID C-KPLY7NUM6X
- **Basic Mechanical Design Course** — MIDO Group, Tec de Monterrey Robotics Program, Jul 2023
- **TOEFL iBT (B2)** — ETS

TECHNICAL SKILLS

CAD:	SolidWorks (4 years, CSWP), Fusion 360, Onshape; parts, assemblies and engineering drawings
Analysis:	Finite Element Analysis (FEA), Computational Fluid Dynamics (CFD), MATLAB
Manufacturing:	Design for Manufacturing (DFM), CNC machining fundamentals, conventional machining, composites, mechanical assembly
Testing:	Prototype development, static fire testing, system integration
Languages:	Spanish (native), English (B2, TOEFL iBT)

ENGINEERING EXPERIENCE

Ignitia Rocket Team, Tecnológico de Monterrey — Present
Design & Manufacturing Team Lead · Board Member

- Lead the team responsible for the design, simulation, manufacturing and validation of experimental rocket components.
- Design mechanical components and assemblies in SolidWorks and Fusion 360; validate them with structural FEA and aerodynamic CFD.
- Design aluminum and composite parts balancing structural performance and manufacturability (DFM).
- Coordinate manufacturing and assembly with propulsion, avionics and recovery; support static fire testing and flight preparation.

International Air and Space Program 2025 — AEXA & NASA, Space Center Houston — Nov 2025
Team 9 · Project MIRKOL · 3rd Place

- Co-designed MIRKOL, a hexapod robot running AI on local servers to give astronauts medical and technical guidance during emergencies on Mars.
- Finished 3rd, one point from 1st place; completed astronaut training and technical sessions with NASA and AEXA engineers.

FIRST Robotics Competition — FRC Daedalus 7102 — 2023 – 2025
Team Captain

- Led multidisciplinary engineering work from concept to competition; coordinated mechanical design and system integration.
- **2025 Mexico National Champion**; qualified for the FIRST World Championship (Houston, TX). **2024 Hermosillo Regional Winner**.

FIRST Tech Challenge — FTC Ikarus 26376 — PrepaTec
Founder & Team Captain

- Founded and managed the team; coordinated engineering activities and technical training for members.
- National Championship qualifier and **Think Award** winner (best-documented engineering process).

AWARDS

- FRC: National Champion (2025) · Hermosillo Regional Winner (2024) · Innovation in Control Award, Regional Finalist (2023)
- Think Award (FTC) · Best Autonomous Robot · Best Control System (2×) · 2nd Place Impact Award (FRC)

LEADERSHIP & OUTREACH

Mentor, FIRST team Sinavolt · Organizer, Guadalajara Off-Season · Founder & President, Kairos Debate Society · Crew Leader, TECHO